

ARPON NAG

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EDUCATION

The University of Hong Kong (HKU)
BEng in Computer Science & BBA in Finance

Sep 2024 - July 2029
Global Engineering and Business Programme Hong Kong

- Recipient of the Hong Kong Jockey Club Chairman's Scholarship (2024) — full-ride award for top 0.1% of international applicants
- Financial Officer, Global Engineering and Business Society, HKU
- Semester Exchange at Uppsala University, Sweden: Completed advanced coursework in Machine Learning, Cyber Security, and Statistical Risk Analysis

EXPERIENCE

eBRAM, HK
Software Engineer Intern

December 2025 - Jan 2026
Central, HK

- Enhanced speech-to-text automation software, improving accuracy and performance by 15%.
- Strategized and reorganized a 1,700+ line codebase, improving documentation clarity and maintainability.
- Designed and implemented new UI and frontend features while coordinating development tasks and managing timelines.

LaunchX Summer Entrepreneurship Program, USA
Applicant Interviewer & Operations Intern

Aug 2024 - Aug 2025
Remote

- Interview and evaluate 100+ applicants for summer programs offered by LaunchX.
- Led and co-led student initiatives, coordinating activities and providing informal guidance to teams developing technology ventures.
- Provide strategic support for planning and executing program expansion across global markets.

University of California, Santa Cruz
Machine Learning Research Intern

May 2023 - Dec 2023
Remote

- Developed cGAN model for medical imaging (98.7% accuracy on 25k histopathology slides)
- Published in IEEE Journal; reduced model inference time by 30%

PROJECTS

Scheduley *2025 - Now*
Intelligent scheduling with AI-assisted planning, calendar integrations, and a React frontend backed by a Node API

- Designed and built an AI-powered scheduling app using TypeScript, React, and Node.js/Express. Implemented multi-model AI fallback (DeepSeek, Claude, Gemini, Azure OpenAI), calendar integrations (Google/Microsoft), Prisma-based data layer, JWT auth, Redis/BullMQ queues, Stripe subscriptions, and production observability with Sentry/OpenTelemetry.

Automatic Anki Deck Generator *2026*
Automatically generate high-quality Anki decks (.apkg) from lecture notes and PDFs

- Built an Automatic Anki Deck Generator in Python that converts lecture notes and PDFs into import-ready .apkg flashcard decks for Anki. Implemented a hybrid generation pipeline using OpenAI (when available) with a robust heuristic fallback, plus OCR support for scanned PDFs. Improved output quality through topic-aware chunking and card validation (deduplication, ambiguity filtering, and formatting constraints). [GitHub Repository](https://github.com/arponnag/anki-deck-generator.git)

Portfolio Risk Analyzer & Optimizer *2025*
Financial Literacy, Risk Assessment, Portfolio Analysis

- A comprehensive Python toolkit for portfolio analysis, risk assessment, and optimization using Modern Portfolio Theory. This project provides tools for calculating risk metrics, optimizing portfolios, performing Monte Carlo simulations, and visualizing results. [GitHub Repository](https://github.com/arponnag/Portfolio-Risk-Analyzer-Optimizer)

TradeSim - Financial Literacy Game *2025*
Financial Literacy, Software Development, Game Development

- TradeSim gamifies financial literacy by letting players experience real-world financial scenarios in a safe, educational environment. Players start at age 22 with \$5,000 and navigate through life events (jobs, emergencies, market crashes) while making investment choices that teach core financial principles. [GitHub Repository](https://github.com/arponnag/TradeSim)

TECHNICAL SKILLS

- **Languages:** Python, Java, C++, JavaScript, R
- **Machine Learning:** Scikit-learn, CNNs, cGANs, Medical Imaging, OpenCV
- **Data Analysis:** Pandas, NumPy, Matplotlib
- **Other:** Git, LaTeX, Frontend Development

NON-TECHNICAL SKILLS

- **Entrepreneurship:** Business development, fundraising, market analysis
- **Research:** Scientific writing, experimental design, statistical analysis, cryptology & data security
- **Leadership:** Team management, project coordination, mentoring

LANGUAGES

- **English:** Fluent **Bengali:** Native
- **Hindi, Urdu, Nepali, Swedish:** Basic